

ROS 2 outer loop

MHE @10 Hz
(m, s, J , confidence;
health/freshness)

/mhe/payload_estimate: m, s, J, γ @10 Hz

NMPC @20 Hz
(acados SQP-RTI)
 $N=20, 1.0$ s

$T, \text{scaled } \omega_{cmd}$ @50 Hz

PX4 inner loop

*motor speeds
+ odometry*

PX4 rate loop
(fixed parameters) @1 kHz

motor cmd

Gazebo physics

Gazebo x500
+ gripper (DART)

*sensing-minimal:
only motor speeds + odometry*